Class 14: RNASeq Mini-Project

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Background

The data for today's mini-project comes from a study of an important HOX gene.

Data Import

```
countData <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
colData <- read.csv("GSE37704_metadata.csv", row.names = 1)
```

Let's have a wee peek at these:

```r
colData
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```r
head(countData)
```

```
                length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Clean up (data tidying)

We need to remove the funny "length" column from our `countData` to make the columns match in the rows in `colData`

```r
countData <- countData[, -1]
```

Check match of `colData` and `countData`

```r
rownames(colData) == colnames(countData)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

```r
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

```r
to.keep <- rowSums(countData) > 0
countData <- countData[to.keep,]
```

DESeq Analysis

```r
library(DESeq2)
```

Setting up the DESeq object

```r
dds <- DESeqDataSetFromMatrix(countData = countData,
                        colData = colData,
                        design = ~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
colData
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

Running DESeq

```r
dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
res <- results(dds)
```

Getting results

```r
res <- results(dds)
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
                      padj
                  <numeric>
ENSG00000279457 6.86555e-01
```

```
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76549e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01
```

Summary of results

```r
res <- results(dds)
summary(res)
```

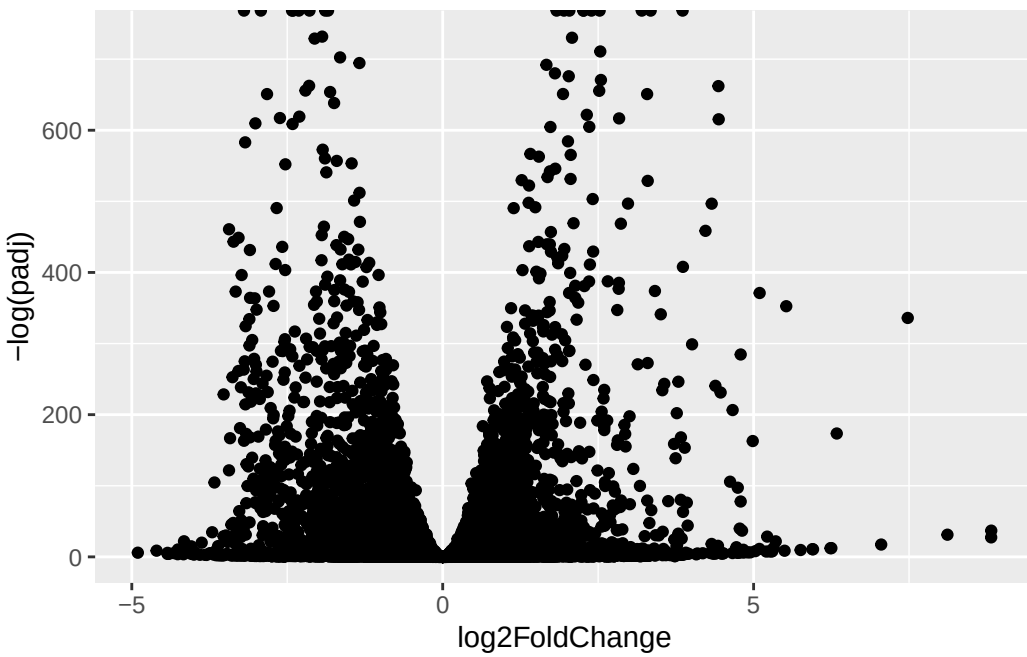
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

A plot of log2 fold change vs -log of Adjusted P-Value

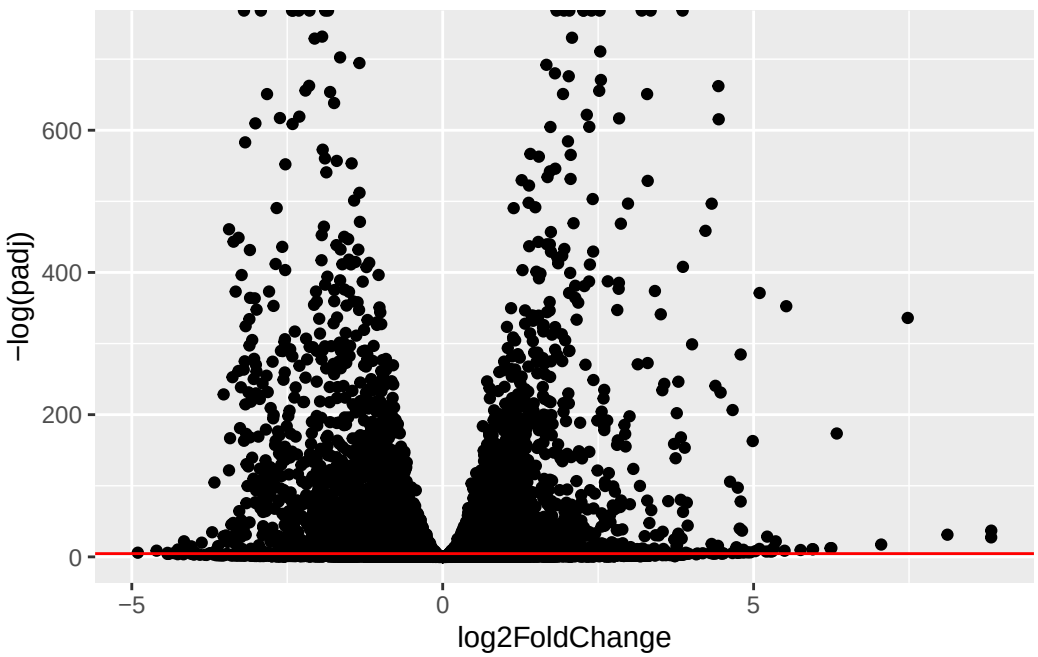
```r
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point() +
  geom_hline(yintercept = -log(0.01), col = "red")
```

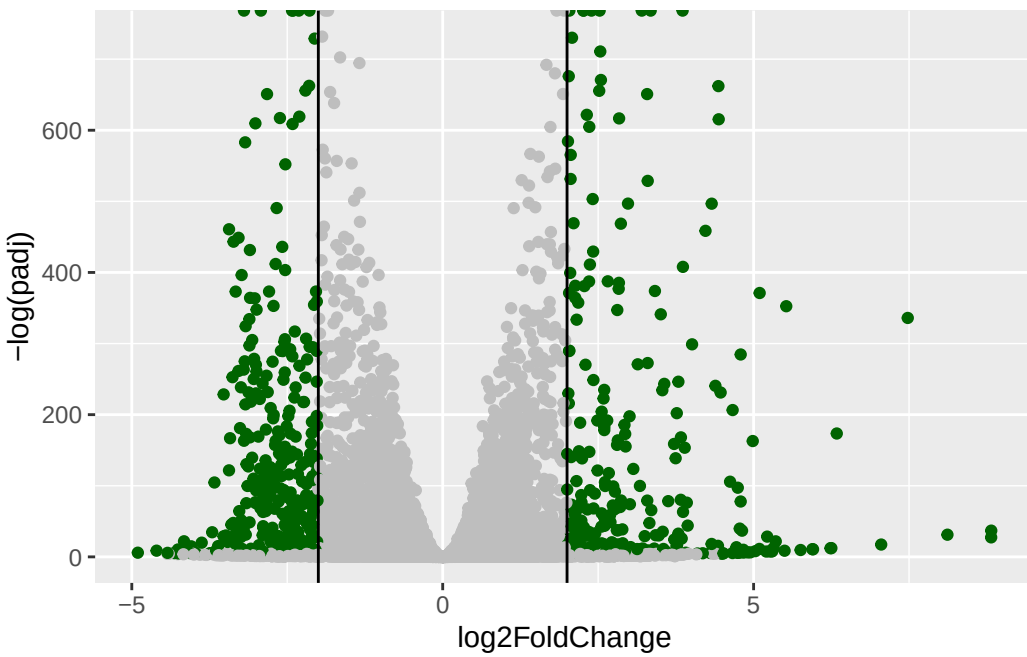
```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Add some color

```
mycols <- rep("gray", nrow(res))
mycols[ abs(res$log2FoldChange) >2 ] <- "darkgreen"
mycols[ res$padj >= 0.01] <- "gray"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col=mycols) +
  geom_vline(xintercept = c(-2,2) )
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
  geom_hline(yintercept = -log(0.01))
```

```
mapping: yintercept = ~yintercept
geom_hline: na.rm = FALSE
stat_identity: na.rm = FALSE
position_identity
```

Add Annotation

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys = rownames(res),
                     keytype = "ENSEMBL",
                     column = "SYMBOL")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = rownames(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 8 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
                      padj      symbol      entrez
                  <numeric> <character> <character>
ENSG00000279457 6.86555e-01          NA          NA
ENSG00000187634 5.15718e-03      SAMD11      148398
ENSG00000188976 1.76549e-35       NOC2L       26155
ENSG00000187961 1.13413e-07      KLHL17      339451
ENSG00000187583 9.19031e-01     PLEKHN1       84069
ENSG00000187642 4.03379e-01       PERM1       84808
```

Pathway Analysis

```r
library(gage)
```

```r
library(gageData)
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

We need a foldchanges vector for `gage()`

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez
```

KEGG

```
data(kegg.sets.hs)
data(sigmet.idx.hs)
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
head(keggres$less)
```

```
                                                   p.geomean stat.mean
hsa04110 Cell cycle                             8.995727e-06 -4.378644
hsa03030 DNA replication                        9.424076e-05 -3.951803
hsa05130 Pathogenic Escherichia coli infection 1.405864e-04 -3.765330
hsa03013 RNA transport                          1.246882e-03 -3.059466
hsa03440 Homologous recombination               3.066756e-03 -2.852899
hsa04114 Oocyte meiosis                         3.784520e-03 -2.698128
                                                       p.val       q.val
hsa04110 Cell cycle                             8.995727e-06 0.001889103
hsa03030 DNA replication                        9.424076e-05 0.009841047
hsa05130 Pathogenic Escherichia coli infection 1.405864e-04 0.009841047
hsa03013 RNA transport                          1.246882e-03 0.065461279
hsa03440 Homologous recombination               3.066756e-03 0.128803765
hsa04114 Oocyte meiosis                         3.784520e-03 0.132458191
                                                set.size         exp1
```

```
hsa04110 Cell cycle                                  121 8.995727e-06
hsa03030 DNA replication                              36 9.424076e-05
hsa05130 Pathogenic Escherichia coli infection        53 1.405864e-04
hsa03013 RNA transport                               144 1.246882e-03
hsa03440 Homologous recombination                     28 3.066756e-03
hsa04114 Oocyte meiosis                              102 3.784520e-03
```

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/areidy/Desktop/BIMM 143/class14

Info: Writing image file hsa04110.pathview.png
```

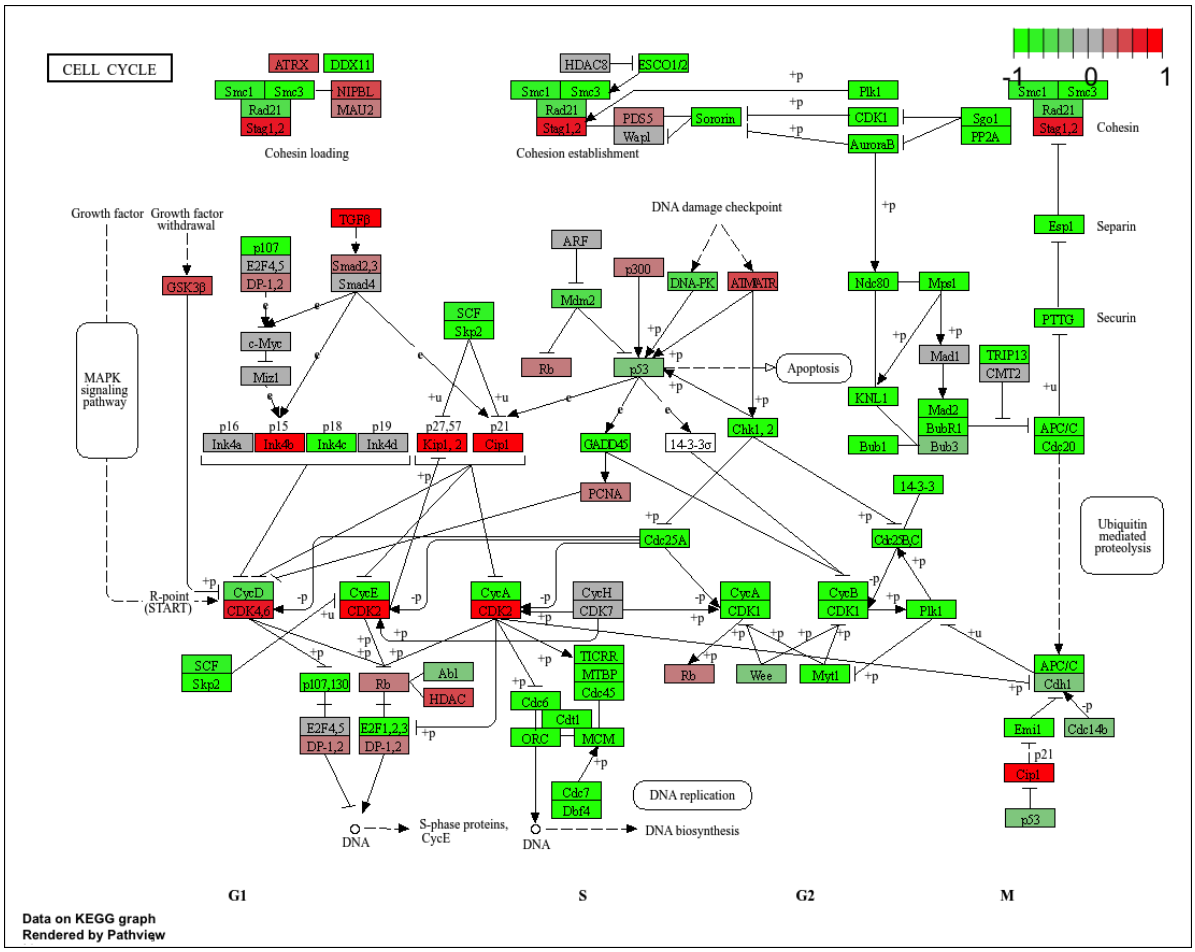


Figure 1: Cell Cycle

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04060" "hsa05323" "hsa05146" "hsa05332" "hsa04640"
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04060")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/areidy/Desktop/BIMM 143/class14
```

```
Info: Writing image file hsa04060.pathview.png
```

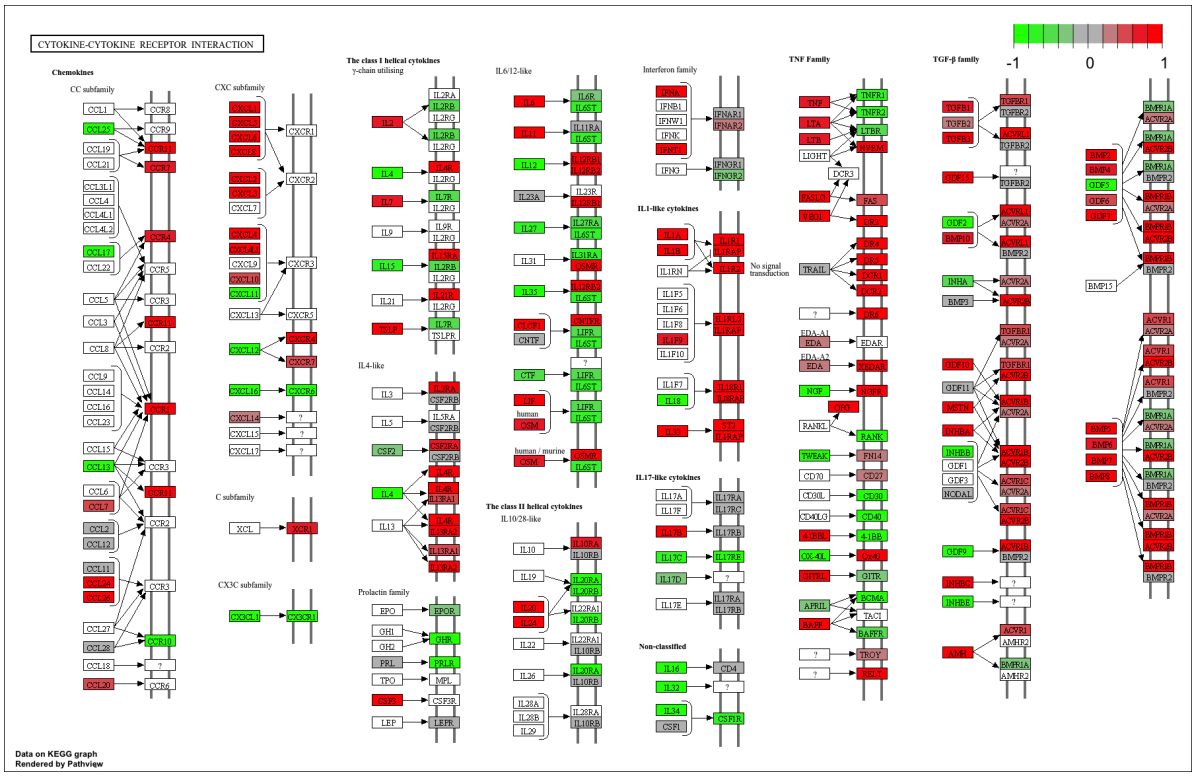


Figure 2: Cytokine-Cytokine Receptor Interaction

```
pathview(gene.data=foldchanges, pathway.id="hsa05323")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/areidy/Desktop/BIMM 143/class14

Info: Writing image file hsa05323.pathview.png
```

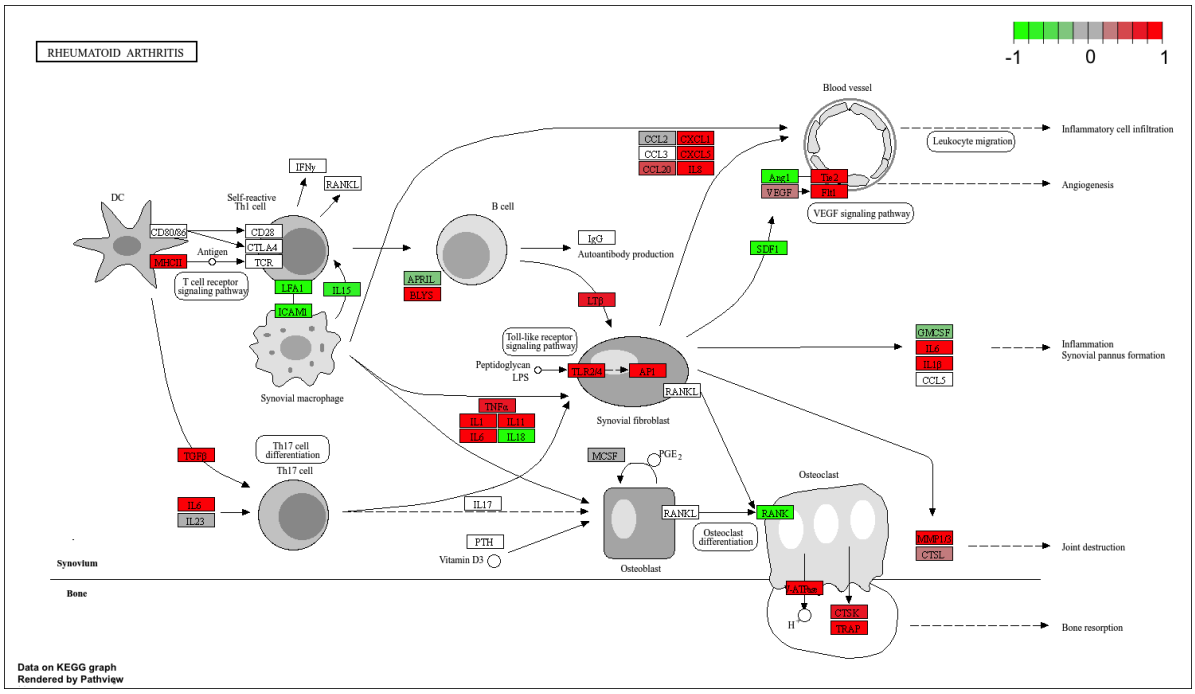


Figure 3: Rheumatoid Arthritis

```
pathview(gene.data=foldchanges, pathway.id="hsa05146")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/areidy/Desktop/BIMM 143/class14
```

```
Info: Writing image file hsa05146.pathview.png
```

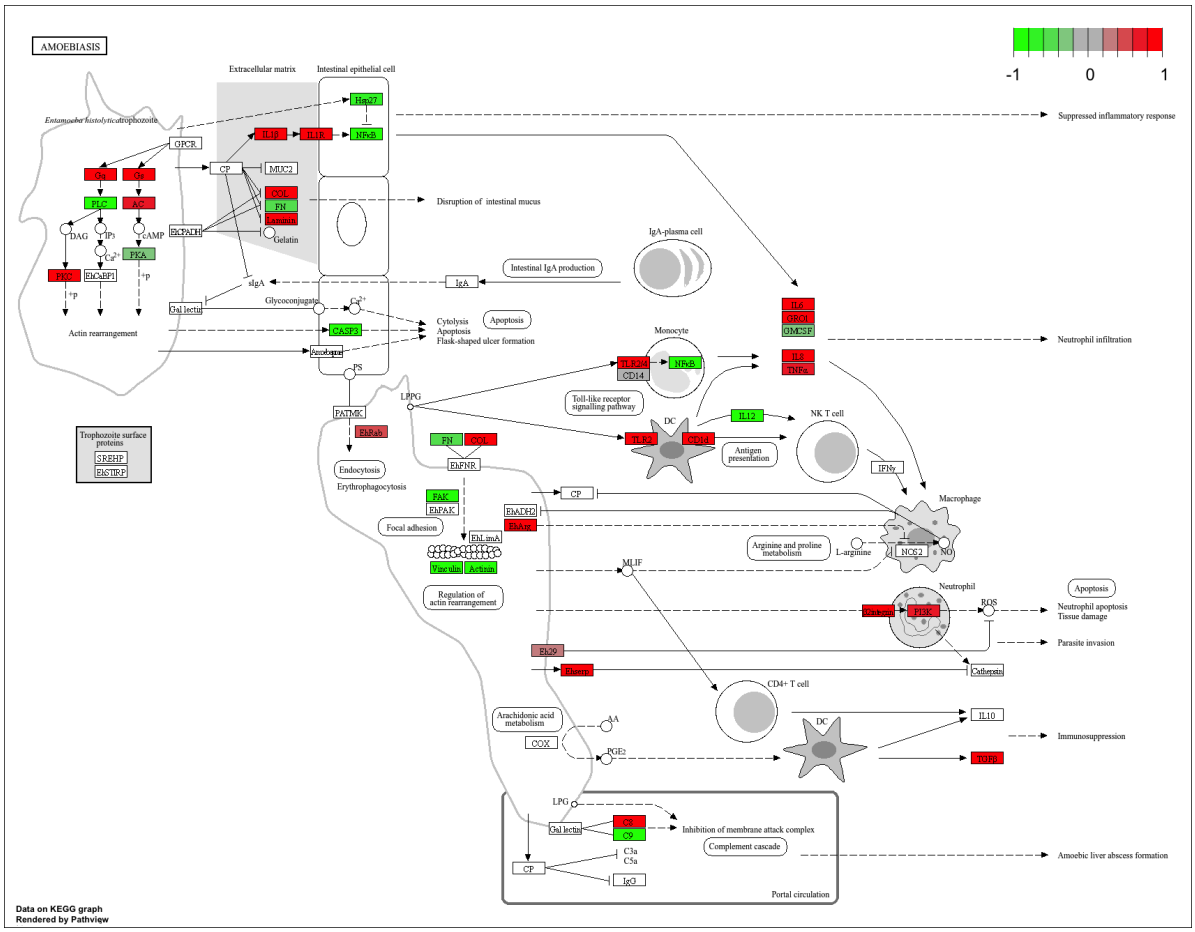


Figure 4: Amoebiasis

```
pathview(gene.data=foldchanges, pathway.id="hsa05332")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/areidy/Desktop/BIMM 143/class14

Info: Writing image file hsa05332.pathview.png
```

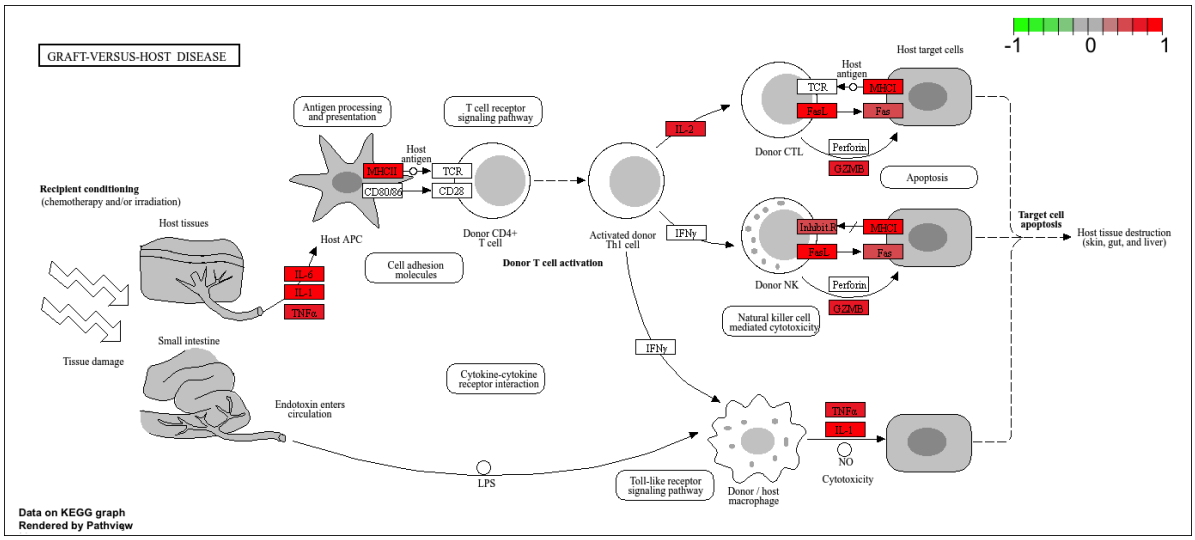


Figure 5: Graft-Versus-Host Disease

```
pathview(gene.data=foldchanges, pathway.id="hsa04640")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/areidy/Desktop/BIMM 143/class14

Info: Writing image file hsa04640.pathview.png
```

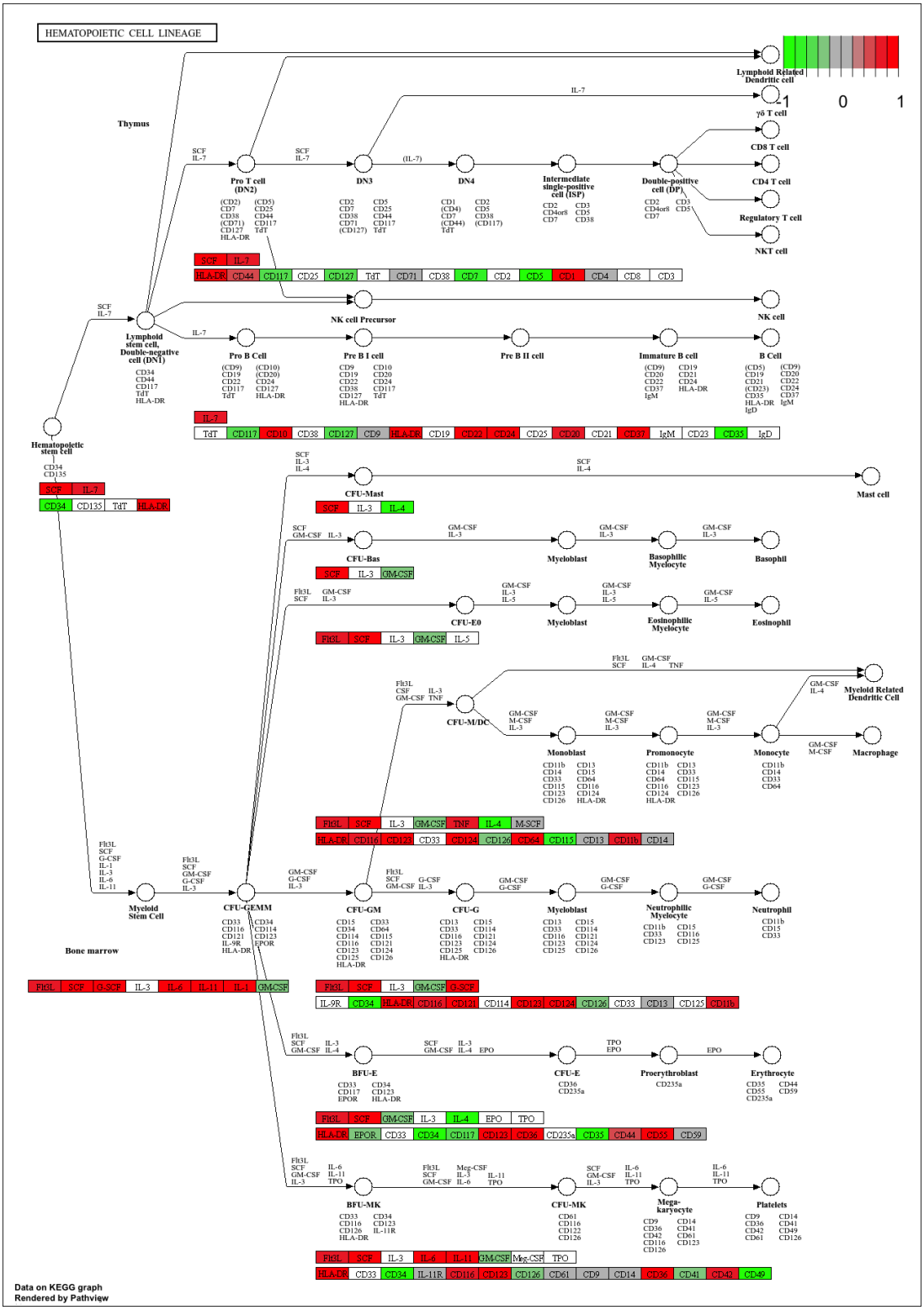


Figure 6: Hematopoietic Cell Lineage

GO

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)
```

```
head(gobpres$less)
```

```
                                          p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.536227e-15 -8.063910 1.536227e-15
GO:0000280 nuclear division              4.286961e-15 -7.939217 4.286961e-15
GO:0007067 mitosis                       4.286961e-15 -7.939217 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.169934e-14 -7.797496 1.169934e-14
GO:0007059 chromosome segregation        2.028624e-11 -6.878340 2.028624e-11
GO:0000236 mitotic prometaphase          1.729553e-10 -6.695966 1.729553e-10
                                                 q.val set.size         exp1
GO:0048285 organelle fission             5.841698e-12      376 1.536227e-15
GO:0000280 nuclear division              5.841698e-12      352 4.286961e-15
GO:0007067 mitosis                       5.841698e-12      352 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.195672e-11      362 1.169934e-14
GO:0007059 chromosome segregation        1.658603e-08      142 2.028624e-11
GO:0000236 mitotic prometaphase          1.178402e-07       84 1.729553e-10
```

Reactome

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

> Q. What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The most significant "entities p-value" is cell cycle, mitotic with a p-value of 2.1E-5. The most significant pathways listed to match KEGG results do not match because they have different pathways that results into more/less genes. These two methods can be differentiated through database sources, pathways, genes, etc.